week-6

$y / (x_i, y_i)$

$$\hat{w}_{ML} = w^* = (XX^T)^{+} X y$$

$$y|_x = w^T x + \epsilon$$

$\epsilon \sim N(0, \sigma^2)$

multiple datasets

$\hat{w}_{ML} =$

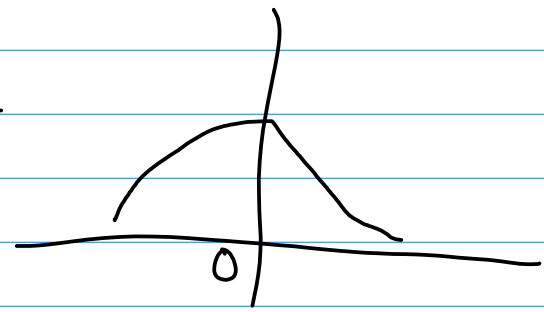
$\hat{w}_{ML} \rightarrow$ MLE $\mathbb{R}^d$

$w \rightarrow$ $\mathbb{R}^d$

$$E\left[\|\hat{w}_{ML} - w\|^2\right]$$

$$= \sigma^2 \cdot \operatorname{trace}\left((XX^T)^{+}\right)$$

$$y_1 = \hat{w}_{ML}^T \cdot x$$

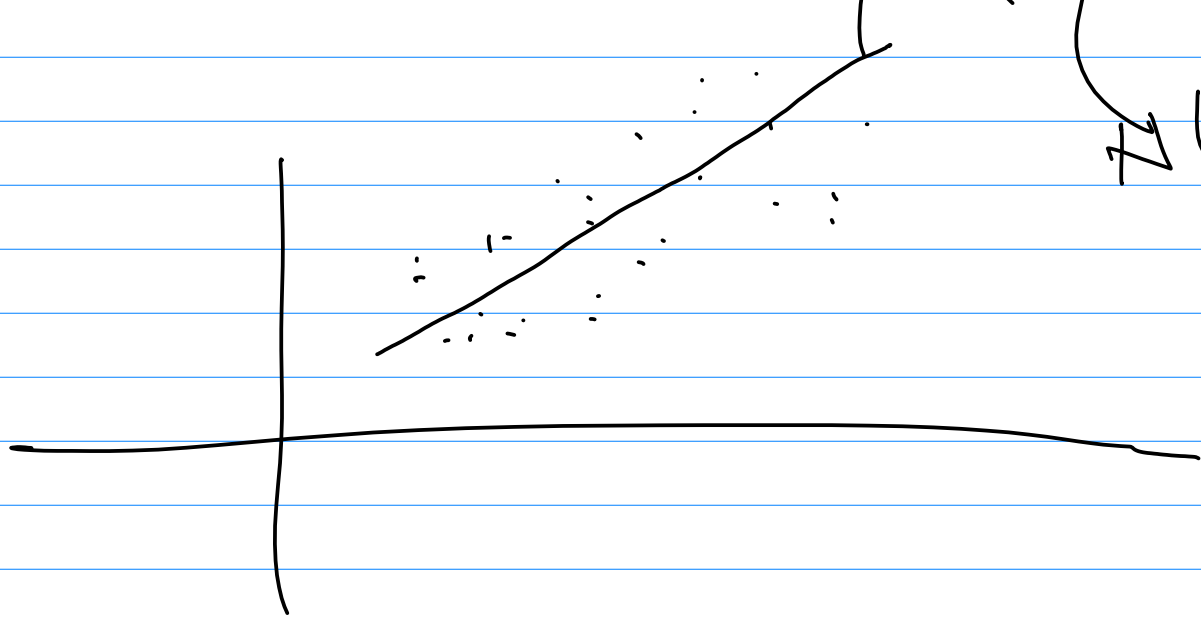


$$\epsilon + w^T x$$

$$N(0, \sigma^2)$$

$$w^T x = 0$$

$$+ \epsilon \to N(0, \sigma^2)$$



$$w^T x + \epsilon \to N(0, \sigma^2)$$

$$(w_0 + w_1 + w_2 \cdots w_n) \begin{cases} 1 \\ x_1 \\ x_2 \end{cases}$$

$$w^T x + b$$

$$\cancel{w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3}$$

$$x_1 = (1)$$

$$(x, y) = \cancel{y = mx}$$

$$y = mx + c$$

$$w_0 + w_1 x.$$

$$E\left[\|\hat{w}_{ML} - w\|^2\right] = \sigma^2 \cdot \underline{\underline{\mathrm{trace}(XX^T)^{+}}}$$

Sum

$$A = \begin{bmatrix} a_1 & & \\ & a_2 & \\ & & a_n \end{bmatrix} = \sum_{i=1}^{n} a_i$$

$$\mathrm{trace}(A) = \sum_{i=1}^{n} a_i = \sum_{i=1}^{n} \lambda_i$$

(XX^T) let's assume the eigenvalue of this matrix to be $\{\lambda_1, \lambda_2 \cdots \lambda_n\}$

$(XX^T)^{-1} \equiv$ eigen value

$$\left\{ \frac{1}{\lambda_1}, \frac{1}{\lambda_2} \cdots \frac{1}{\lambda_n} \right\}$$

$$E\left(\|\hat{w}_{ML} - w\|^2\right) = \sigma^2 \cdot \left(\sum_{i=1}^{n} \left(\frac{1}{\lambda_i} \right) \right)$$

Squared error:

training

$\hat{w}_{ML} = w^* =$

$y_1 = \hat{w}_{mL}$

multiple dataset

$\hat{w}_{mL} =$

$$\hat{w}_{new} = (XX^T + \lambda I)^{-1} X y$$

$\in \mathbb{R}^{n \times n}$

$\in \mathbb{R}$

$Av_1 = \lambda_1 v_1$

$(A + \lambda I) v_1 = Av_1 + \lambda I v_1$

$= Av_1 + \lambda v_1$

$$= \lambda_1 v_1 + \lambda v_1$$

$$(A + \lambda I) v_1 = (\lambda_1 + \lambda) v_1$$

Let matrix A, we are calculating eigen, eigenvectors of A.

$$(XX^T + \lambda I)$$

$$A v_1 = \lambda_1 v_1$$

$$A \rightarrow \boxed{A + \lambda I} \rightarrow$$

$$(A + \lambda I) v_1 = A v_1 + \lambda I v_1$$

$$= A v_1 + \lambda v_1$$

$$= \lambda_1 + v_1 + \lambda v_1$$

$$= v_1 (\lambda_1 + \lambda)$$

$$(A + \lambda I) v_1 = \underline{\underline{(\lambda_1 + \lambda)}} v_1$$

$$X X^T \rightarrow (X X^T + \lambda I)$$

$$\lambda_1, \ldots$$

$$\sum_{i=1}^{n} \frac{1}{\lambda_i + \lambda}$$

Existence

$\exists \ \lambda \in \mathbb{R}$ s.t.

$$\hat{w}_{new} = (X X^T + \lambda I)^{-1} X y$$

lesser error over the $\underline{\underline{\hat{w}_{m2}}}$

cross - validation.

training set validation set

80%	20

d.

K-Fold cross-validation

F_1	F_2	F_3	...	F_K

$K = 5$

$F_1 - F_4$ F_5

$F_1, F_2, F_3, F_5 \rightarrow F_4$

F_3

F_2

F_1

$K = 3$

F_1, F_2, F_3

$F_1 F_2 \rightarrow F_3$

$F_2 F_3 \rightarrow F_1$

$F_1 F_3 \rightarrow$

100

30 F

30 F_2

90 F_3

$$\hat{w}_{new} = (XX^T + \lambda I)^{-1} X Y$$

$=$

$($

$(x_1, y_1)\ (x_2, y_2) \cdots (x_n, y_n)$

$$\boxed{y|(\text{Data}) \sim w^T x_1 + \epsilon}$$

$$y_{pred} = \hat{w}_{new}^T x_{test}$$

$x_1 \to y_1$

$x_2 \to y_2$

$x_3 \to y_3$

training

$(x_1, y_1$

$w \cdot x + c = y.$

$(x_1, y_1), (x_2, y_2) \cdots (x_n, y_n)$

$x_1 - y_1$

$x_2 \quad y_2$

$\vdots \quad \vdots$

$x_n - y_n$

$y_i \simeq \underline{\underline{w^T x_i}} + \epsilon \quad \sim N(0, \sigma^2)$

$y_1 = \underline{\underline{w_1^T x_1 + \epsilon}}$

$\hat{w}_{ML}$

L.R.

$$\sum_{i=1} \left(w_{ML}^T x_i - y_i \right)^2$$

$$\left(w^T x_i + \epsilon \right)$$

$$\left(w_{ML}^T x_i - w^T x_i + \epsilon \right)^2$$

$$\operatorname{tr}\left(XX^T\right)^{-1} \cdot \sigma^2$$

$$\sigma^2 \cdot \operatorname{tr}\left(V^{-1}\right)$$

$$\lambda =$$

$$\hat{w}_{new} = \left(XX^T + \lambda I \right)^{-1} X y$$

70 - point

$$\boxed{w_{new} = (X X^T + \lambda I)^{-1} X y}$$

Bayesian Estimation of view of Linear Regression.

posterior(w | data) = prior(w) × Likelihood

prior(P)

| data

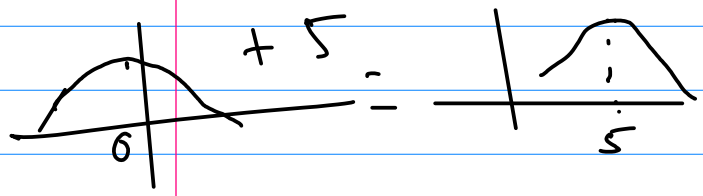
posterior(P / data)

Posterior(w / Data) $\sim$ Likelihood($\frac{D}{w}$) · Prior(w)

$x_1 \cdots x_n$

$y_1 \cdots y_n$

$(x_1, y_1), (x_2, y_2) \cdots (x_n, y_n)$



(x_{n+1}, y_{n+1})

y_i / $y = w^T x + \epsilon_{N(0,\sigma^2)} \sim$

$$\frac{y_i}{x} \approx N(w^T x, \sigma^2)$$

$$N(w^T x, 1)$$

$\{w.$

$$w \sim N(\overset{0 \in \mathbb{R}^d}{0}, \nu^2 I)$$

$$\text{prior}(w) \sim N(0, \nu^2 I)$$

$w \in \mathbb{R}^d$ $\qquad$ $d \times d$

Posterior $\qquad$ Likelyhood $\qquad$ prior

$$P(w / (x_1, y_1) \ldots (x_n, y_n)) \propto p((x_1, y_1) \ldots (x_n, y_n) / w) \cdot p(w)$$

$$\propto \left(\prod_{i=1}^{n} \frac{e^{-\frac{(y_i - w^T x_i)^2}{2}}}{\sqrt{2\pi}} \right) \cdot \prod_{i=1}^{d} \frac{e^{-\frac{(w_i - 0)^2}{2\nu^2}}}{\sqrt{2\pi}}$$

$$\propto \left(\prod_{i=1}^{n} e^{-\frac{(y_i - w^T x_i)^2}{2}} \right) \cdot e^{-\sum_{i=1}^{d} \frac{w_i^2}{2\sigma^2}}$$

$$\frac{\sum_{i=1}^{d} w_i^2}{2\sigma^2} = \frac{\|w\|^2}{\sigma^2}$$

$$w = [w_1 \cdots w_d]$$

$$\|w\| = w_1^2 + w_2^2 \cdots w_d^2$$

$$\propto \prod_{i=1}^{n} e^{-\frac{(y_i - w^T x_i)^2}{2}} \cdot e^{-\frac{\|w\|^2}{2\sigma^2}}$$

<u>MAP</u>

take log on Both side

$$\log(\text{posterior}) = \log\left(\prod_{i=1}^{n} e^{-(y_i}\right.$$

$$= \sum_{i=1}^{n} -\frac{(y_i - w^T x_i)^2}{2} - \frac{\|w\|^2}{2\sigma^2}$$

$$\hat{w}_{MAP} = \underset{w}{\arg\max}$$

$$= \underset{w}{\arg\min} \sum_{i=1}^{n} \frac{(y_i - w^T x_i)^2}{2} + \frac{\|w\|^2}{2\sigma^2}$$

$$\nabla f(w) = (XX^T)w - Xy + \frac{w}{\sigma^2}$$

$$\hat{w}_{MAP} = \left(XX^T + \frac{1}{\sigma^2} I\right)^{-1} Xy$$

$$w_{new} = (XX^T + \lambda I)^{-1} Xy$$

MAP estimation of LR

with gaussian prior $N(0, \sigma^2 I)$.

it is equivalent

$$\hat{w}_R = \underbrace{\sum_{i=1}^{n} \frac{(y_i - w^T x_i)^2}{2}}_{\text{error}^2} + \lambda \|w\|^2$$

Regularizer

Ridge error

$w_1 \cdots w_d$

$x_1 \cdots x_d$

$$X = \begin{matrix} x_1 & x_2 & \cdots & x_d \end{matrix}$$

$$\begin{array}{c} \qquad\qquad\qquad\qquad t \qquad\qquad\qquad\qquad y \\ X \left[\begin{array}{c|c|c} \text{height} & \text{weight} & 2\,\text{height} + 3\,\text{weight} \\ 4 & 3 & 0 \end{array}\right] = \begin{array}{c} 4\,\text{height} + 3\,\text{weight} \\ \\ \end{array} \end{array}$$

Ridge: $\|w\|^2$

0 3 4

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